BILINEARITY, SKEW SYMMETRY AND THE TWO NORMALIZATION PROPERTIES OF THE TAME SYMBOL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The preliminary preparation of the tame symbol in a finite extension. Same chapter chow.tex, label lemma-prepare-tame-symbol, invoked at proof lines 913 and 933. Produces, for the finitely many nonzerodivisors involved, a finite ring extension in which the elements become monomials in a single nonzerodivisor up to units, which reduces each verification to an identity among the units and the exponents.

Source reference. Original equation 0EAQ: sources/stacks-chow/pinned-original-source.tex, source lines 820–820. Navigation pointer for \ref{equation-tame-symbol} used inside the counted span (line 903); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAK: sources/stacks-chow/pinned-original-source.tex, source lines 779–779. Navigation pointer for \ref{item-1-x} used inside the counted span (line 907); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAP: sources/stacks-chow/pinned-original-source.tex, source lines 794–794. Navigation pointer for \ref{item-1-x-better} used inside the counted span (line 905); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAI: sources/stacks-chow/pinned-original-source.tex, source lines 773–773. Navigation pointer for \ref{item-bilinear} used

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inside the counted span (line 907); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAL: sources/stacks-chow/pinned-original-source.tex, source lines 782–782. Navigation pointer for \ref{item-bilinear-better} used inside the counted span (line 904); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAN: sources/stacks-chow/pinned-original-source.tex, source lines 790–790. Navigation pointer for \ref{item-normalization} used inside the counted span (line 905); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAJ: sources/stacks-chow/pinned-original-source.tex, source lines 776–776. Navigation pointer for \ref{item-skew} used inside the counted span (line 907); the argument that consumes it is retained in the span.

Source reference. Original reference 0EAM: sources/stacks-chow/pinned-original-source.tex, source lines 786–786. Navigation pointer for \ref{item-skew-better} used inside the counted span (line 904); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EAG: sources/stacks-chow/pinned-original-source.tex, source lines 642–642. Navigation pointer for \ref{lemma-prepare-tame-symbol} used inside the counted span (line 913); the argument that consumes it is retained in the span.

Lemma 0.1. *The tame symbol (0EAQ) satisfies (0EAL), (0EAM), (0EAN), (0EAP) and hence gives a map $\partial_A : Q(A)^* \times Q(A)^* \to \kappa(\mathfrak{m})^*$ satisfying (0EAI), (0EAJ), (0EAK).*

Proof. Let us prove (0EAL). Let $a_1, a_2, a_3 \in A$ be nonzerodivisors. Choose $A \subset B$ as in Lemma 0EAG for a_1, a_2, a_3. Then the equality

$$\partial_A(a_1a_2, a_3) = \partial_A(a_1, a_3)\partial_A(a_2, a_3)$$

follows from the equality

$$(-1)^{(e_{1,j}+e_{2,j})e_{3,j}}(u_{1,j}u_{2,j})^{e_{3,j}}u_{3,j}^{-e_{1,j}-e_{2,j}} = (-1)^{e_{1,j}e_{3,j}}u_{1,j}^{e_{3,j}}u_{3,j}^{-e_{1,j}}(-1)^{e_{2,j}e_{3,j}}u_{2,j}^{e_{3,j}}u_{3,j}^{-e_{2,j}}$$

in B_j. Properties (0EAM) and (0EAN) are equally immediate.

Let us prove (0EAP). Let $a_1, a_2, a_1 - a_2 \in A$ be nonzerodivisors and set $a_3 = a_1 - a_2$. Choose $A \subset B$ as in Lemma 0EAG for a_1, a_2, a_3. Then it suffices to show

$$(-1)^{e_{1,j}e_{2,j}+e_{1,j}e_{3,j}+e_{2,j}e_{3,j}+e_{2,j}}u_{1,j}^{e_{2,j}-e_{3,j}}u_{2,j}^{e_{3,j}-e_{1,j}}u_{3,j}^{e_{1,j}-e_{2,j}} \bmod \mathfrak{m}_j = 1$$

This is clear if $e_{1,j} = e_{2,j} = e_{3,j}$. Say $e_{1,j} > e_{2,j}$. Then we see that $e_{3,j} = e_{2,j}$ because $a_3 = a_1 - a_2$ and we see that $u_{3,j}$ has the same residue class as $-u_{2,j}$. Hence the formula is true – the signs work out as well and this verification is the reason for the choice of signs in (0EAQ). The other cases are handled in exactly the same manner. □